

Report 08

Geospatial Intelligence: Bathymetry, Route Optimisation & Passage Planning

Project:	JalJeev — Agentic Marine Intelligence System
Phase:	Phase 0 (Advanced), Landing Ahead of Phase 2 Schedule
Focus:	GEBCO bathymetry, H3 grid routing, radial zone scan, port-to-port passage

1. Overview

This report covers three interconnected geospatial capabilities that were originally planned for Phase 1–2 but were built during Phase 0: (1) the GEBCO regional bathymetry tiling system, (2) the Route Agent's radial candidate scan for fishermen, and (3) the Passage Agent's port-to-port corridor planner for sailors and commercial vessels.

2. GEBCO Bathymetry — The Zero-Fill Disaster

2.1 The Problem

The first attempt to download GEBCO 2026 bathymetry data tried to fetch an India-wide subset ($\sim 3^\circ \times 5^\circ$) in a single OPeNDAP request from the CEDA THREDDS server.

Result: The download appeared to succeed — a valid NetCDF file was saved — but **96% of the depth values past the first ~200 rows were silently zero-filled**. This was not an error; the OPeNDAP protocol had silently truncated the response.

Consequence: Every downstream depth lookup returned "0 m / land" for almost everywhere offshore. The Geo Agent's grounding check incorrectly hard-vetoed almost every ocean point.

2.2 The Fix: Regional Tiling with Sanity Checks

The bathymetry adapter was redesigned to:

- Fetch small $\sim 1^\circ \times 1^\circ$ regional tiles (one per demo area)
- Run an explicit **>85% nonzero-fraction** sanity check before saving
- A failed tile is reported and NOT saved
- Points outside all loaded tiles return status: "unavailable"

2.3 Loaded Tiles (as of Aug 27, 2026)

Tile Name	Lat	Lon	Purpose
kochi	9.30–10.30	75.30–76.30	Arabian Sea demo
chennai	12.60–13.60	79.80–80.80	Coromandel demo
vizag	17.20–18.20	82.90–83.90	Primary Andhra demo point
puri	19.30–20.30	85.30–86.30	Odisha original
paradip	19.80–20.80	86.30–87.30	Mahanadi mouth
puri_offshore	18.80–19.80	85.30–86.30	Deep water south of Puri
odisha_se_offshore	19.00–20.00	86.30–87.30	SE offshore
gopalpur	18.80–19.80	84.40–85.40	Chilika / Gopalpur

Contiguous Odisha coverage: **18.8–20.8 N, 84.4–87.3 E.**

2.4 The Puri Shallow Water Correction

Project notes had recorded that "0 m depth extends tens of km offshore near the Mahanadi delta," which was initially used to rule Odisha out as a demo region.

Measured against the loaded tiles, the transect due south of Puri at 85.83 E reads:

- **11 km offshore:** 26 m depth
- **22 km offshore:** 16 m depth
- **33 km offshore:** 413 m depth

Good working water. The earlier note conflated the Puri coastline (correctly 0 m and land-vetoed) with the water offshore. Odisha was reinstated as a viable demo region.

3. Route Agent: Radial Candidate Scan

3.1 The Fisherman's Question

"Where near me is it safe?"

This is a point query with no destination. The Route Agent answers it by scanning outward:

3.2 Algorithm

1. Take the user's position (or a specified departure point)
2. Generate **16 candidate zones** in a radial pattern at compass headings (N, NNE, NE, ...)
3. At multiple distances (10, 20, 30, 40 km) along each heading
4. For each candidate: fetch weather + ocean data, run hard-constraint check, compute risk score
5. Rank survivors by risk score (lowest = safest)
6. Return the top 3 safest candidates with full evidence

3.3 Key Feature: Route Hazards, Not Just Destination Scores

The Route Agent doesn't just score the destination — it scores intermediate waypoints too. A recommendation that is safe at both ends but **dangerous in the middle** is caught.

"50% along route: water depth 0.0 m is less than the vessel's draft (0.6 m) plus a 1.0 m safety margin — grounding risk."

4. Route Optimizer: A* on H3 Risk Grid

4.1 Beyond Naive Candidate Scan

While the candidate scan finds "which direction is best," the Route Optimizer finds the **optimal path** to the safest cell within a radius.

4.2 Implementation

- **Grid:** H3 hexagonal grid at resolution 5 (~5 km cells)
- **Graph:** Each H3 cell is a node; edges connect neighbouring cells (k-ring = 1)
- **Edge weights:** Risk score of the destination cell
- **Algorithm:** A* search (via NetworkX) from origin to the best candidate
- **Output:** Sequence of H3 cells representing the optimised route, total risk along the path

5. Passage Agent: Port-to-Port Planning

5.1 The Sailor's/Trader's Question

"Get me from Kochi to Kakinada and tell me when I arrive."

This is a completely different problem from the fisherman's radial scan:

5.2 Algorithm

1. Resolve departure and arrival ports from the ports table
2. Compute great-circle track between the two ports
3. Build an **H3 corridor** along the track, dilated laterally so A* can route around hazards
4. Adaptive H3 resolution by passage distance
5. For each leg: score against the **forecast hour the vessel actually arrives there**

5.3 The Tacking Convergence Fix (Aug 28, 2026)

For sailing vessels, the passage ETA depends on the point of sail at each leg. But the point of sail depends on the wind direction at the ETA hour. This is circular.

Fix: Iterative convergence — re-run the ETA calculation until the change between iterations is < 15 minutes. Converges in 2–3 iterations for typical passages.

5.4 Forecast Horizon Honesty

Anything past the 7-day Open-Meteo horizon is flagged `beyond_forecast_horizon`, never silently filled in. A 10-day passage gets detailed weather scoring for the first 7 days and an explicit "beyond forecast" caveat for the remaining 3.

6. Best Verified Demo Points

Location	Coordinates	Depth	Notes
Offshore Visakhapatnam	17.65, 83.35	~40–56 m	Best demo point — steep shelf, genuinely non-vetoed
Alternate Vizag	17.60, 83.45	~40 m	Also good
Offshore Puri	19.70, 85.83	~26 m @ 11 km	Viable — the city itself is correctly vetoed
Bhubaneswar	20.30, 85.82	Land	Intentional veto demo — 60 km inland

7. Deliverables

Item	Planned Phase	Delivered In
GEBCO bathymetry tiling	Phase 1	■ Phase 0
Route Agent (radial scan)	Phase 1	■ Phase 0
Route Optimizer (A* on H3)	Phase 2	■ Phase 0
Passage Agent (port-to-port)	Phase 2	■ Phase 0

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